WORKED SOLUTION

STEP II 2009, Question 10

Mechanics • Collisions and Restitution

STEP II 2009, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s2-q10

The key idea

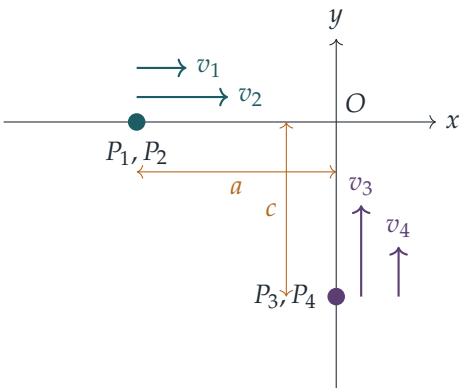
There are four collisions, but only the first two need any mechanics: they fix the four speeds just afterwards. Everything else is timing. Two particles that "collide at O" while travelling along different axes simply reach O at the same instant, so each later collision gives one equation of the form distance/speed = distance/speed. Divide one timing equation by the other and the unknown distances drop out, leaving a condition on the masses.

Solution

Setting up

We model the four objects as particles moving on smooth axes, so between collisions each moves with constant velocity. Let P_2 start at distance a from O on the negative x-axis and P_3 at distance c from O on the negative y-axis. We must show $a = c$.

Take $t = 0$ to be the moment of the first two (simultaneous) collisions. These happen where P_2 and P_3 are sitting, so at $t = 0$ the particles P_1 and P_2 are both at distance a from O, and P_3 and P_4 are both at distance c from O.



The first two collisions

Suppose a particle of mass M moving with speed u hits a stationary particle of mass m directly, with coefficient of restitution e, and afterwards they have velocities V and w in the original direction of motion. Conservation of momentum and Newton's experimental law give

$$Mu = MV + mw, \qquad w - V = eu,$$

so

$$w = \frac{M(1+e)u}{M+m}, \qquad V = \frac{(M-em)u}{M+m}.$$

Applying this to P_1 hitting P_2, and to P_4 hitting P_3, the velocities towards O just after $t = 0$ are

$$v_1 = \frac{(m_1 - em_2)u}{m_1+m_2}, \qquad v_2 = \frac{m_1(1+e)u}{m_1+m_2},$$
$$v_4 = \frac{(m_4 - em_3)u}{m_3+m_4}, \qquad v_3 = \frac{m_4(1+e)u}{m_3+m_4}.$$

Clearly $v_2, v_3 > 0$. Since P_1 and P_4 do later reach O, they must still be moving towards it, so $v_1 > 0$ and $v_4 > 0$ too.

Using the later collisions

Ask yourself How can two particles on perpendicular axes "collide"?

The only point the two axes share is O. So "P_2 and P_3 collide at O" just means they arrive at O at the same time, and likewise for P_1 and P_4. Between $t = 0$ and their arrival at O each particle moves at constant speed (the question tells us the next collision any of them is involved in happens at O), so the arrival times are distance over speed:

$$\frac{a}{v_2} = \frac{c}{v_3} \qquad (P_2 \text{ and } P_3 \text{ meet at } O), \tag{1}$$
$$\frac{a}{v_1} = \frac{c}{v_4} \qquad (P_1 \text{ and } P_4 \text{ meet at } O). \tag{2}$$

Dividing (1) by (2), the distances cancel:

$$\frac{v_1}{v_2} = \frac{v_4}{v_3}.$$

From the formulae above, the factors u and $m_1 + m_2$ (or $m_3 + m_4$) cancel in each ratio, so this says

$$\frac{m_1 - em_2}{m_1(1+e)} = \frac{m_4 - em_3}{m_4(1+e)},$$

and multiplying through by $m_1 m_4(1+e)$,

$$m_4(m_1 - em_2) = m_1(m_4 - em_3), \qquad \text{so} \qquad e\,m_2 m_4 = e\,m_1 m_3.$$

Since $e > 0$ we may divide by e, so $m_1 m_3 = m_2 m_4$, that is,

$$\frac{m_1}{m_2} = \frac{m_4}{m_3} = k, \text{ say.}$$

Now go back to v_2 and v_3. Dividing top and bottom by m_2 and m_3 respectively,

$$v_2 = \frac{k(1+e)u}{k+1} = v_3.$$

So P_2 and P_3 leave the first collisions with equal speeds, and then (1) gives $a = c$. Hence P_2 and P_3 were initially equidistant from O, as required.

Checks

- Take $m_1 = 2$, $m_2 = 1$, $m_3 = 3$, $m_4 = 6$ (so $k = 2$), $e = \frac{1}{2}$, $u = 1$ and $a = c = 1.7$. Then $v_2 = v_3 = 1$ and $v_1 = v_4 = \frac{1}{2}$: P_2 and P_3 reach O together at $t = 1.7$, and P_1 and P_4 together at $t = 3.4$, later as the question says. A step-by-step simulation agrees.
- Note $v_2 - v_1 = eu > 0$, so P_1 really does arrive after P_2, consistent with "some time later".
- If $e = 0$ the argument breaks, as it should: then $v_1 = v_2$ and $v_3 = v_4$, so (2) is the same as (1), and any a and c with $a/c = v_2/v_3$ work. That is exactly why the question insists $e > 0$.

Common slips

- Measuring P_1's journey from its starting point. After $t = 0$ it starts from where P_2 was, at distance a from O, which is why the same a appears in both (1) and (2).
- Dividing by e without comment. The case $e = 0$ really is different.
- Trying to analyse the collision at O between P_2 and P_3. Nothing about its outcome is needed; only the fact that they get there together.

Beyond STEP

The mass condition $m_1 m_3 = m_2 m_4$ says the two collisions are scaled copies of each other: velocities after a direct impact depend only on the mass ratio and e, a first glimpse of dimensional analysis and similarity. At university, collisions are cleanest in the centre-of-mass frame, where the coefficient of restitution simply scales the relative velocity. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops momentum and collisions in this way.

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